

Non-Technical Setup & Batch Engine Guide

How to batch-generate 365 high-resolution infographics, customize topics, and run the pipeline without technical knowledge.

⚡ The Core Feature: Automated Batch Generation

Instead of opening design software and clicking "Export" 365 separate times, the batch engine renders every single day sequentially in a headless background browser. In just 2–3 minutes, it generates a full year of production-ready, 2400x1350 PNG infographics into your output folder with a single command.

1. Overview: How It Works

This system works like an automated printing press on your computer. You do not need Canva, Photoshop, or paid subscriptions. It separates design from content:

📁 The Ingredients (`data/all_days.json`)

A plain text file where your 365 subjects, top apps, SWOT points, and summaries live.

🎨 The Stamp (`templates/layout.html`)

The visual template with colors, cards, and branding. It automatically adjusts to fit your words.

⚙️ The Batch Printer (`render.js`)

A free headless browser (Puppeteer) that renders each day in memory and snaps a 4K image.

🖼️ The Output (`output/`)

Where your finished files arrive: `day_001.png` through `day_365.png`.

2. How the Batch Engine Operates

The batch generator includes three built-in safeguards designed for non-technical users:

- **Smart Auto-Resume:** If you stop the engine halfway through (or shut down your computer), it checks what is already in `output/` and skips files that exist. It only generates what's missing.
- **Single-Day Testing vs. Full Batch:** In `render.js`, changing `i < 1` lets you test a single graphic in 2 seconds before running all 365 with `i < allDays.length`.
- **Zero-Cost Scaling:** Whether generating 1 graphic or 3,650 graphics, there are zero API fees, cloud server costs, or rendering limits.

3. The Code Files (Copy & Paste)

Create these files in your folder using **Notepad**. Save each file with the exact name shown.

File 1: `package.json` (Save in `E:\Infographics\package.json`)

```
{
  "name": "3sci-infographics-engine",
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "render": "node render.js",
    "build:data": "node build_all_days.js",
    "build:manual": "node build_manual_pdf.js"
  },
  "dependencies": {
    "ejs": "^3.1.9",
    "puppeteer": "^22.0.0"
  }
}
```

File 2: `render.js` (The Batch Engine)

```
import fs from 'fs';
import path from 'path';
import puppeteer from 'puppeteer';
import ejs from 'ejs';

async function run() {
  const templateStr = fs.readFileSync('./templates/layout.html', 'utf-8');
  const allDays = JSON.parse(fs.readFileSync('./data/all_days.json', 'utf-8'));

  let logoBase64 = '';
  if (fs.existsSync('./templates/logo_white.png')) {
    logoBase64 = `data:image/png;base64,${fs.readFileSync('./templates/logo_white.png').toString('base64')}`;
  }

  const browser = await puppeteer.launch({ headless: 'new', args: ['--no-sandbox'] });
  const page = await browser.newPage();
  await page.setViewport({ width: 2400, height: 1350 });

  console.log(`Starting batch render for ${allDays.length} topics...`);

  // BATCH LOOP: Renders all 365 days in sequence automatically
  for (let i = 0; i < allDays.length; i++) {
    const dayData = allDays[i];
    dayData.day_number = i + 1;
    dayData.logo_src = logoBase64;
    const outputPath = path.join('./output', `day_${String(i + 1).padStart(3, '0')}.png`);

    // Auto-skip files already created
    if (fs.existsSync(outputPath)) {
      console.log(`[Skip] Day ${i + 1} already exists.`);
      continue;
    }

    const html = ejs.render(templateStr, dayData);
    await page.setContent(html, { waitUntil: 'domcontentloaded' });
    await page.screenshot({ path: outputPath, type: 'png' });
    console.log(`[Rendered ${i + 1}/${allDays.length}] Day ${i + 1}: ${dayData.title}`);
  }

  await browser.close();
  console.log('Batch complete! All images created in output folder.');
```

```
}  
run();
```

File 3: `templates/layout.html`

Save your master graphic design file inside `E:\Infographics\templates\layout.html`. It controls fonts, dark backgrounds, high-contrast badges, SWOT panels, and automatic text sizing.

4. How to Change Parameters & Subject Matter

You never need to edit computer code to change your content. All information lives in `data/all_days.json`. You can open and edit this file with **Notepad**.

STEP 1

Open the Data File in Notepad

Right-click `data/all_days.json` and choose **Open with > Notepad**.

STEP 2

Find the Entry You Want to Change

Press `Ctrl + F` and type `"day_number": 1`. You will see clearly labeled sections:

```
{
  "day_number": 1,
  "title": "YOUR MAIN HEADLINE HERE",
  "sector": "Your Industry Category",
  "overview": "Write a 2-3 sentence summary of the subject matter here.",
  "top_15": [
    { "name": "App or Company Name", "role": "What it does", "strength": "Key advantage", "friction": "Key challenge" }
  ],
  "adoption_metrics": [
    { "label": "Key Area 1", "val": 85 },
    { "label": "Key Area 2", "val": 60 }
  ],
  "swot": {
    "strengths": ["Clear strength bullet 1", "Clear strength bullet 2"],
    "weaknesses": ["Clear weakness bullet 1"],
    "opportunities": ["Clear opportunity bullet 1"],
    "threats": ["Clear threat bullet 1"]
  },
  "summary": "Your final one-sentence executive verdict."
}
```

STEP 3

Save and Execute Batch Render

Press `Ctrl + S` to save the file. In your terminal, type:

```
npm run render
```

The batch engine will check all days, render any new or updated entries, and place them directly in the `output` folder.

✨ Always ask your preferred AI platform if you need help updating code!

You never have to figure out code or formatting errors by yourself. If you want to add new topics, switch color themes, or create fresh data, simply copy the text of your file and paste it into ChatGPT, Claude, or Gemini with this prompt:

"Here is my data file. Please keep the exact same format and structure, but rewrite Day [X] to cover [Your New Topic]."

Your AI will format the entire block of code cleanly for you to copy and paste.

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